Mock C

125 min · 44 marks · 26 MCQ + 9 short answer · open book, no internet or AI. Circle one option per MCQ; tick **guessed** if unsure.

Section A: Multiple choice (1 mark each)

Question 1 guessed ☐ [1 mark]

What is a key benefit of using cloud-based databases compared to traditional on-premise non-cloud databases?

- **(a)** Increased security
- **(b)** Lower upfront costs
- **(c)** Increased accessibility and scalability
- **(d)** Cheaper storage
- **(e)** Lower latency

Question 2 guessed ☐ [1 mark]

A company adds two identical web servers by hand every Friday at 5 pm, because traffic always rises at weekends, and removes them on Monday morning. Which description fits best?

- **(a)** Elastic scaling
- **(b)** Horizontal scaling, done manually
- **(c)** Vertical scaling
- **(d)** Cloud bursting
- **(e)** The elastic resource capacity architecture

Question 3 guessed ☐ [1 mark]

A company manages development, staging, and production environments using Terraform. Which one of the following approaches best avoids duplication and ensures consistency?

- **(a)** Copy-paste configurations per environment
- **(b)** Use separate state files
- **(c)** Use only one environment
- **(d)** Hardcode environment values
- **(e)** Keep separate code repositories

Question 4 guessed ☐ [1 mark]

Your company uses virtual machines (VMs) and is experiencing resource contention (e.g., CPU, memory, disk I/O, or network). Which solution is most effective to address this issue comprehensively?

- **(a)** Add more physical servers
- **(b)** Add more memory
- **(c)** Use faster storage devices
- **(d)** Increase your network speeds
- **(e)** Optimise VM resource allocation

Question 5 guessed ☐ [1 mark]

A team clusters social-media users by follower count using k-means, after taking log2 of every count. Two of the final centroids are 4 (cluster P) and 10 (cluster Q). How does a typical user in Q compare with a typical user in P?

- **(a)** About 6 times as many followers
- **(b)** About 2.5 times as many followers
- **(c)** About 64 times as many followers
- **(d)** About 1,024 times as many followers
- **(e)** About 100 times as many followers

Question 6 guessed ☐ [1 mark]

A company wants to move its deployment to the cloud but is worried about the security threats in the cloud. Which one of the following is the major security threat that the company needs to worry about?

- **(a)** Unauthorized physical access
- **(b)** Data breaches due to multi-tenancy
- **(c)** Outdated antivirus software
- **(d)** Limited user authentication methods
- **(e)** Data leaks due to bad programming practices

Question 7 guessed ☐ [1 mark]

A company, GreatestCloudOfAll Inc, have noticed that their cloud infrastructure frequently scales up and down, which is causing great instability and hence extra costs. You have been called in as an expert to advise them. What would your advice be?

- **(a)** You just need to increase your number of instances and it'll be fine
- **(b)** You can re-check your thresholds for scaling and also add a cooldown period which prevents further scaling during this period
- **(c)** You can monitor it and over time just disable scaling as that will help in the long run in this case
- **(d)** Best is manual scaling, nothing beats it!
- **(e)** Reduce the overall cost by reducing your number of instances

Question 8 guessed ☐ [1 mark]

Your company operates a multi-cloud environment (e.g., AWS and Azure) and is experiencing data synchronization issues between the clouds, such as incomplete transfers or data loss. What is the most likely cause?

- **(a)** Slow storage performance on one cloud
- **(b)** Limited virtualisation on one cloud
- **(c)** Incompatible cloud APIs
- **(d)** Uneven resource provisioning (e.g., storage, compute, or network)
- **(e)** Inconsistent security configurations

Question 9 guessed ☐ [1 mark]

A hospital H uses a SaaS patient-records system run by provider P. Which statement is correct?

- **(a)** H must extend its trust boundary to cover P, but P does not need to extend its trust boundary to cover H
- **(b)** P must extend its trust boundary to cover H, because P stores H's data
- **(c)** Once the service is shared, neither H nor P has an organisational boundary
- **(d)** H and P each extend their trust boundary to cover the other
- **(e)** The trust boundaries of H and P never overlap, because each has its own organisational boundary

Question 10 guessed ☐ [1 mark]

You wish to build your own application and are operating in a highly competitive market where being first is more important than being perfect, and rapid iteration is critical. Which cloud service model is most appropriate?

- **(a)** IaaS
- **(b)** FaaS
- **(c)** CaaS
- **(d)** PaaS
- **(e)** SaaS

Question 11 guessed ☐ [1 mark]

Your company has built a SaaS that is meant to serve around 5000 customers. However, your SaaS has become very popular (yay!) and suddenly there is a demand from about 25000 users. This will lead to an increase in resource utilisation. What are the incremental maintenance costs (mainly management overheads such as configuring servers, patching, etc.) of adding these new resources?

- **(a)** About the same as before
- **(b)** Twice the original cost
- **(c)** Three times the original cost
- **(d)** Four times the original cost
- **(e)** Five times the original cost

Question 12 guessed ☐ [1 mark]

A team's Terraform configuration declares 3 web servers, and its state file lists 3. An engineer then deletes one of the servers by hand in the cloud console. What happens the next time someone runs terraform apply?

- **(a)** Nothing: the state file still says 3, so Terraform believes everything matches
- **(b)** Terraform destroys the other two servers and recreates all three
- **(c)** Terraform stops with an error until someone edits the state file by hand
- **(d)** Terraform looks up each server in the state against the real cloud, finds that only 2 exist, and creates one to match the configuration
- **(e)** Terraform rewrites the configuration so that it declares 2 servers

Question 13 guessed ☐ [1 mark]

When expecting a huge variation in traffic, as a cloud engineer, we need to be able to ensure the application can scale dynamically based on demand. Which cloud computing service model would we use to implement scalability?

- **(a)** Elastic load balancer
- **(b)** SaaS
- **(c)** Auto-scaling
- **(d)** PaaS
- **(e)** None of the above

Question 14 guessed ☐ [1 mark]

Which of the following are signs of a highly available application?

(i) Applications are protected behind multiple layers of security

(ii) Spikes in user demand are met through automatically increasing resources

(iii) A failure in one geographic region will trigger an automatic switch to resources in a different region

(iv) Virtualized hypervisor-driven systems are deployed as mandated by company policy

(v) Application developers ensure the necessary software updates are applied

- **(a)** (i) and (iii)
- **(b)** (ii) and (iv)
- **(c)** Only (v)
- **(d)** (ii) and (iii)
- **(e)** (i) and (v)

Question 15 guessed ☐ [1 mark]

What is a primary challenge when migrating legacy on-premise systems to the cloud?

- **(a)** Decreased performance
- **(b)** Increased data compliance
- **(c)** Increased management
- **(d)** Increased costs
- **(e)** Application compatibility

Question 16 guessed ☐ [1 mark]

A k-means job on customers' yearly spend produces one cluster holding a single customer who spent $2 million, while the other four clusters are almost identical. The team takes the logarithm of spend and reruns the job. Why does this help?

- **(a)** It makes the data smaller, so the job finishes faster
- **(b)** It removes the $2 million customer from the data set
- **(c)** It automatically raises k, so more clusters are formed
- **(d)** It makes every customer's spend equal so the clusters match
- **(e)** It pulls extreme values closer to the rest, so one outlier no longer claims a whole cluster and the other customers can separate into meaningful groups

Question 17 guessed ☐ [1 mark]

Your company has a policy of encrypting all data that is stored. Your data is hosted in a virtual private cloud and attached to it are multiple storage devices which has the data that must be encrypted. You have the following options available. Based on this, which one of the following options will allow you to do this? (if multiple options are possible choose the one that allows you the most choice)

(i) Ask clients to encrypt their data before they give it to you

(ii) Use third party encryption tools to encrypt the data

(iii) Use native encryption tools available in the Operating System to encrypt the data

(iv) Encrypt your data within the application before storing

(a) (i) or (iii)

(b) Only (iv)

(c) (ii) or (iii) or (iv)

(d) (i) or (ii) or (iii)

(e) (iii) or (iv)

Question 18 guessed ☐ [1 mark]

What is the advantage of a single tenancy cloud environment over a multitenancy environment?

(a) Lower costs due to shared resources

(b) Enhanced data security and isolation

(c) Faster scalability for user spikes

(d) Simplified resource management

(e) Reduced need for compliance controls

Question 19 guessed ☐ [1 mark]

Your company has recently gotten to grips with Terraform and wish to use it. However, they also wish to have automatic drift detection during continuous integration (the practice of merging code changes into a central repository frequently). Which of the following is the best approach to achieve this?

(a) Run terraform apply daily

(b) Disable drift checks

(c) Use manual audits

(d) Periodically recreate the infrastructure

(e) Run terraform plan and provide an alert on changes

Question 20 guessed ☐ [1 mark]

A web fleet runs behind a load balancer, and the whole fleet is defined in infrastructure code. At 3 a.m. one server crashes. Which pieces actually notice the failure and bring a replacement?

(a) The Terraform configuration file, which keeps watching the servers

(b) The Terraform state file, which marks the dead server as failed

(c) The Git repository holding the code, which triggers a rebuild

(d) The terraform plan command, which runs continuously in the background

(e) The load balancer's health check, and an auto-scaling group that launches a fresh server from the same definition

Question 21 guessed ☐ [1 mark]

You are part of an e-commerce start-up and wish to understand the shopping patterns arising from last year's sales events. In order to do this, you have been tasked with doing the data analysis where you have a lot of data. Which cloud deployment will you choose as the best option?

- **(a)** Public cloud only
- **(b)** Private cloud only
- **(c)** Hybrid cloud
- **(d)** Either Public or Private cloud
- **(e)** Multi-cloud

Question 22 guessed ☐ [1 mark]

Which level of cloud service is usually provided for the application to run the development environment?

- **(a)** PaaS
- **(b)** IaaS
- **(c)** SaaS
- **(d)** FaaS
- **(e)** None of the above

Question 23 guessed ☐ [1 mark]

A company wishes to migrate its legacy application to the cloud quickly with minimal changes. Which approach from the following would be most suited (you can assume that all options listed are technically feasible)?

- **(a)** Refactor the application to use serverless functions and then deploy it
- **(b)** Get a cloud provider to build a SaaS equivalent solution for you and then use it
- **(c)** Rewrite the application using a PaaS platform which allows for easy deployment
- **(d)** Use IaaS and move application as is from on-premise to cloud
- **(e)** Keep it on-premise

Question 24 guessed ☐ [1 mark]

A company defines its network and servers once in a Terraform module. Dev, staging and production all call this module, but all three are kept in one folder that shares a single state file. What is the main risk?

- **(a)** An apply run to change dev also covers the staging and production resources, so a dev change can alter production
- **(b)** A module cannot be reused by more than one environment
- **(c)** Terraform can no longer create resources in parallel
- **(d)** The three environments will slowly drift apart because each has its own copy of the code
- **(e)** Terraform cannot run plan until the three environments are split into separate repositories

Question 25 guessed ☐ [1 mark]

You are the founder of a start-up with very limited personnel. Your company handles the input and processing of infectious disease clinical trial data on the cloud. The data processing needs to be real-time. The data contain highly sensitive, personally identifiable information. However, you want to reduce as much manual work as you can due to your small team. Which service should you choose for this application?

(a) IaaS
(b) FaaS
(c) CaaS
(d) PaaS
(e) SaaS

Question 26 guessed ☐ [1 mark]

Which one of the following statements is NOT true about Cloud Computing?

(a) Opex includes expenses like electricity and maintenance for on-premise IT resources
(b) Detecting spam in social media is a good example of a use case for MapReduce
(c) PaaS is the cloud service that is usually provided to run the development environment
(d) Elastic scaling can dynamically increase or decrease server resources based on a specific period
(e) Higher security threats are a disadvantage of using public PaaS for SaaS development

Section B: Short answer (2 marks each)

Question 27 [2 marks]

You are a Solutions Architect at a leading technology company. The IT division of the metropolitan MRT (Mass Rapid Transit) system has approached you to design a next-generation payment platform to support SimplyGo and phase out the legacy EZ-Link card system. A key concern is that during peak hours, the current system experiences high latency, resulting in long queues and poor commuter experience. The reason for this is that for each user, the system performs fare calculation, balance deduction, and displays the update immediately before letting the user pass through the gate.

Your task is to propose two ways in which this payment platform can be scalable and also have low-latency.

Question 28 [2 marks]

During an outage an engineer opens port 8080 on a firewall by hand in the cloud console. The firewall is managed by Terraform, and the code still has that port closed. A week later a teammate runs terraform apply to resize a database. In two or three sentences: what happens to port 8080, why, and what should the engineer have done?

Question 29 [2 marks]

The company you work for is responsible for a large web server farm. You are tasked with increasing the availability of this farm to ensure little to no downtime. List two ways in which you can achieve this.

Question 30 [2 marks]

It is the upcoming holiday season and your application has gained a lot of traction. However, the downside is that your cloud infrastructure is struggling with this load. You have looked into auto-scaling and load-balancing already. List two other methods that you can use to help with this issue.

Question 31 [2 marks]

An online exam platform is used by universities in Singapore, Malaysia and Indonesia. Each country's law says student records must stay inside that country. Demand is about 50 times higher during two-week exam periods and close to zero otherwise. Give two design choices that cover both the residency rules and the demand pattern.

Question 32 [2 marks]

Your workplace wishes to make the work environment a bit more fun. They are going to provide advanced collaboration services, such as video, chat, and an online whiteboard. Given the cost of hosting the infrastructure for this, they have decided that a cloud-based approach is the way to go. Which cloud computing deployment model would best suit your company's needs? Explain your answer.

Question 33 [2 marks]

In a video-sharing app, upload web servers and video encoders run as separate groups. A user uploads a 2 GB video and instantly sees "upload complete", but the video reaches viewers 20 minutes later. A colleague calls this a bug and wants each upload server to wait for the encoding to finish before replying to the user. In two or three sentences, explain why the current design is sensible and what the colleague's change would cause.

Question 34 [2 marks]

Assume that you are responsible for organising your cloud resources globally as the leader of an enterprise. Discuss at least two key factors that you should consider when deciding on cloud resource hosting.

Question 35 [2 marks]

You are a cloud developer who is deploying your application (which has a steady-state workload) to AWS. Which pricing option would you choose and why?